

Programming Assignment 4

Running the project

Run `src\CTP\Main.java` and provide it with the input file as the first argument.

I am using the same syntax from the assignment, with an added command that specify the numbers of simulation to perform on the given input.

Syntax:

`#C 10` ; *Running 10 simulations on the input*

Default value (if not given) is 5.

States

Denote **n** the number of vertices in the graph, and **m** the number of fragile edges.

States are a possible combination of fragile edge's status and player current location.

i.e. a Cartesian product:

$$\{V_1, V_2, \dots, V_n\} \times \{FREE, BLOCKED, UNKNOWN\}^m$$

The utility of each state signal how "far" it is from a goal state. While each goal state (i.e. a state at the location of the package deliver destination) have utility 0.

Note that this is a minimization problem.

Evaluating states

The program calculates the utility of each state based on the **value iteration** algorithm taught in class.

The algorithm start with a utility of 0 for each state, and in each iteration of the algorithm we update the utility values to make the locally consistent with the **Bellman Equation**.

We stop once the maximum change in an iteration is smaller than a pre-defined value $\delta = 0.25$.

Simulating travel

In each simulation, each fragile edge get a concrete value (FREE / BLOCLED) based on its provided probability from the input file.

The Agent start at the vertex k specified in the input file by the command $\#A V_k$, with all the fragile edges status being UNKNOWN to him. i.e. the initial state is:

$$V_k \times UNKNOWN^m$$

And the player follows optimal policy, from each state attempting to go to the state with the lowest utility value.

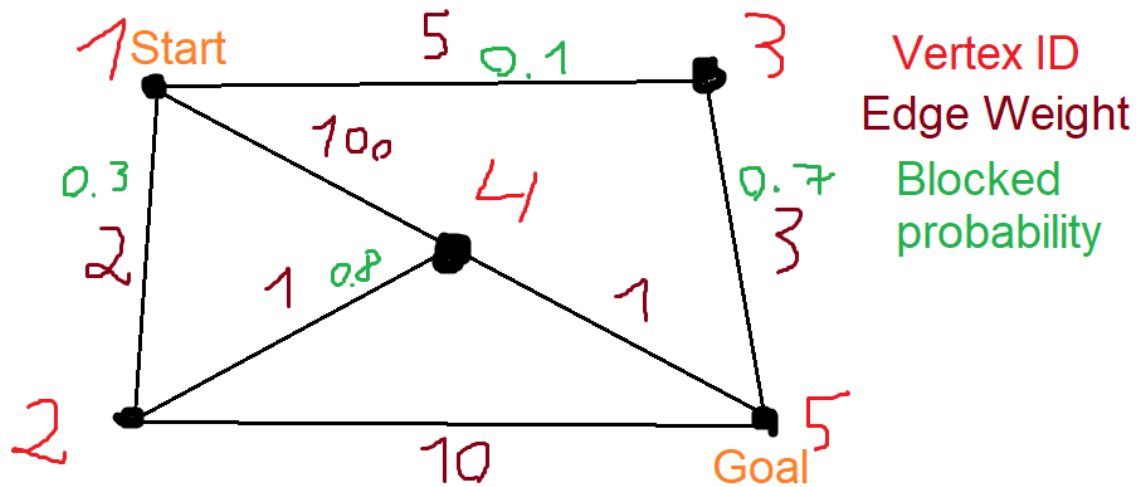
Upon encountering a fragile edge, either it is found to be free, and than it traverse it, moving to a different vertex and updating and status for said edge to be free.

Or it turns out the fragile edge is blocked, so it stays in the same vertex, and update the status of said fragile edge to be blocked.

The simulation stop when reaching a goal state, or a state which is impossible to reach goal state from.

- Below there's 2 running examples of different scenarios. The output log is quite large. Example 1 start at page 3, and example 2 start at page 63.

Running example 1:



Input File:

#N 5

#E1 1 2 W2

#E2 1 3 W5

#E3 1 4 W100

#E4 2 4 W1

#E5 2 5 W10

#E6 3 5 W3

#E7 4 5 W1

#P1 D V5 ; Package 1 to be delivered at V5 (already carried)

#F V1 V2 0.3 ; Edge from 1 to 2 is fragile, blocked with probability 0.3

#F V1 V3 0.1

#F V2 V4 0.8

#F V3 V5 0.7

#A V1 ; Agent starts at V1

#C 10 ; Running 10 simulations on the input

Game logs:

starting...

===== MDP Belief-state: =====

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 8.0

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 12.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 12.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

[Conf
ident
ial]

Utility score: 8.0

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 12.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 12.0

[Conf
ident
ial]

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 12.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 4.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 4.0

Best Action: 2

[Conf
ident
ial]

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 12.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 12.0

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.24

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 10.4

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 8.0

Best Action: 3

[Conf
ident
ial]

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 80.1

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 8.0

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 80.1

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.0

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 80.1

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 101.0

Best Action: 4

Location: V1

[Conf
ident
ial]

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 17.3

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 82.19

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 17.3

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

[Conf
ident
ial]

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 82.19

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 17.3

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 101.0

Best Action: 4

Location: V1

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 82.19

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 5.199999999999999

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 33.099999999999994

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 26.83

[Conf
ident
ial]

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 8.0

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 38.699999999999996

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 32.429999999999999

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.0

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 37.58

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 31.309999999999995

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 33.099999999999994

Best Action: 2

[Conf
ident
ial]

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 33.099999999999994

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 33.099999999999994

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 38.699999999999996

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 38.699999999999996

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 38.699999999999996

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 37.58

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 37.58

Best Action: 2

[Conf
ident
ial]

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 37.58

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 7.99

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 33.099999999999994

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 27.457

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 11.07

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 38.699999999999996

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 33.057

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 10.958

Best Action: 3

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 37.58

Best Action: 2

Location: V1

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 31.936999999999998

Best Action: 2

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 1

Location: V2

[Conf
ident
ial]

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

[Conf
ident
ial]

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

[Conf
ident
ial]

Best Action: 4

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{FREE}, (V3,V5) = \text{BLOCKED})$

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{FREE}, (V3,V5) = \text{UNKNOWN})$

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{BLOCKED}, (V3,V5) = \text{FREE})$

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{BLOCKED}, (V3,V5) = \text{BLOCKED})$

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{BLOCKED}, (V3,V5) = \text{UNKNOWN})$

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{UNKNOWN}, (V3,V5) = \text{FREE})$

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: $((V1,V2) = \text{FREE}, (V1,V3) = \text{UNKNOWN}, (V2,V4) = \text{UNKNOWN}, (V3,V5) = \text{BLOCKED})$

Utility score: 8.4

Best Action: 4

[Conf
ident
ial]

Location: V2

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

[Conf
ident
ial]

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

[Conf
ident
ial]

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

[Conf
ident
ial]

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 1

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

[Conf
ident
ial]

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

[Conf
ident
ial]

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 2.0

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 10.0

Best Action: 5

[Conf
ident
ial]

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 10.0

Best Action: 5

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 8.4

Best Action: 4

Location: V2

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 8.4

Best Action: 4

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 9.0

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 7.2

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 17.0

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 12.799999999999999

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 15.4

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 11.68

Best Action: 5

Location: V3

[Conf
ident
ial]

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Unreachable state

[Conf
ident
ial]

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 106.0

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 75.1

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

[Conf
ident
ial]

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 106.0

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 75.1

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 106.0

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 75.1

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

[Conf
ident
ial]

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

[Conf
ident
ial]

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)
Utility score: 1.258385194403621E307
Best Action: 1

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)
Utility score: 3.0
Best Action: 5

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)
Utility score: 38.099999999999994
Best Action: 1

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)
Utility score: 27.569999999999993
Best Action: 5

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)
Utility score: 3.0
Best Action: 5

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)
Utility score: 43.699999999999996
Best Action: 1

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)
Utility score: 31.489999999999995
Best Action: 5

Location: V3
Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

[Conf
ident
ial]

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 42.58

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 30.705999999999996

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Unreachable state

[Conf
ident
ial]

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Unreachable state

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.2583851944036209E308

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

[Conf
ident
ial]

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 3.0

Best Action: 5

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.7976931348623158E307

Best Action: 1

Location: V3

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.258385194403621E307

Best Action: 1

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

[Conf
ident
ial]

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

[Conf
ident
ial]

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

[Conf
ident
ial]

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

[Conf
ident
ial]

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

[Conf
ident
ial]

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

[Conf
ident
ial]

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

[Conf
ident
ial]

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

[Conf
ident
ial]

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 1.0

Best Action: 5

Location: V4

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 1.0

Best Action: 5

Location: V4

[Conf
ident
ial]

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 1.0

Best Action: 5

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

[Conf
ident
ial]

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

[Conf
ident
ial]

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

[Conf
ident
ial]

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = FREE ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

[Conf
ident
ial]

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

[Conf
ident
ial]

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

[Conf
ident
ial]

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = BLOCKED ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

[Conf
ident
ial]

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = FREE ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

[Conf
ident
ial]

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

[Conf
ident
ial]

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = BLOCKED ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = FREE ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = BLOCKED ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V2) = UNKNOWN ,(V1,V3) = UNKNOWN ,(V2,V4) = UNKNOWN ,(V3,V5) = UNKNOWN)

Utility score: 0.0

Best Action: 0

===== Running a simulation #1 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 3 Best action: V1, fragile edge. It appears to be free, walking on it.

Agent location: 1 Best action: V4, non-fragile edge. walking on it.

Agent location: 4 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #2 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #3 =====

[Conf
ident
ial]

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #4 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #5 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #6 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #7 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be free, walking on it.

Agent location: 4 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #8 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #9 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #10 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V2, fragile edge. It appears to be free, walking on it.

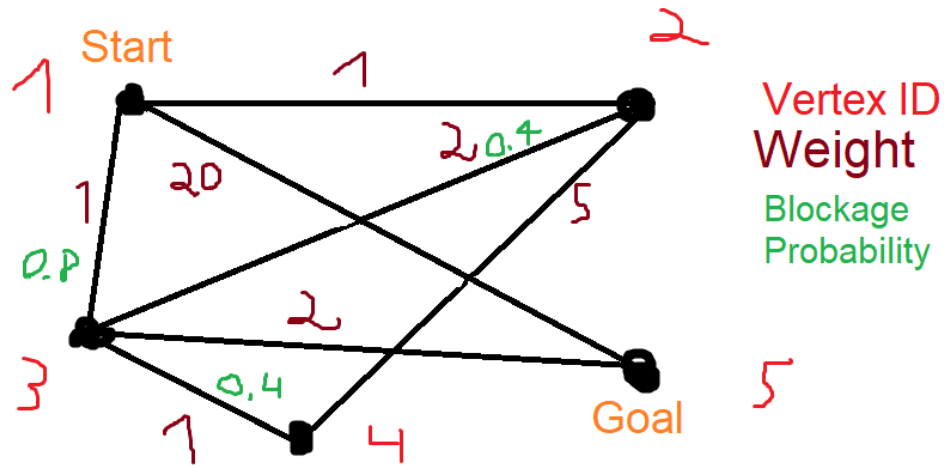
Agent location: 2 Best action: V4, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

Finished the simulations... The program will now exit.

Running example 2:



Input File:

#N 5

#E1 1 2 W1

#E2 1 3 W1

#E3 1 5 W20

#E4 2 3 W2

#E5 2 4 W5

#E6 3 4 W1

#E7 3 5 W2

#P1 D V5 ; Package 1 to be delivered at V5 (already carried)

#F V1 V3 0.8 ; Edge from 1 to 3 is fragile, blocked with probability 0.8

#F V2 V3 0.4

#F V3 V4 0.4

#A V1 ; Agent starts at V1

#C 10 ; Running 10 simulations on the input

Game logs:

starting...

===== MDP Belief-state: =====

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

[Conf
ident
ial]

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 3.0

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 5.0

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 5.0

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 5.0

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 9.0

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 20.0

[Conf
ident
ial]

Best Action: 5

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 18.2

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 6.6

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 11.8

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 10.28

Best Action: 2

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 4.6

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 4.6

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 4.6

Best Action: 3

[Conf
ident
ial]

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 7.8

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 16.6

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 15.16

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 5.88

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 10.040000000000001

Best Action: 3

Location: V1

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 8.824

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 4.0

Best Action: 1

[Conf
ident
ial]

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 4.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 4.0

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 4.0

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 4.0

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 8.0

Best Action: 4

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 21.0

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 17.2

Best Action: 4

Location: V2

[Conf
ident
ial]

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 5.6

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 10.8

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 9.28

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 4.0

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 4.0

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 4.0

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 8.0

Best Action: 4

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

[Conf
ident
ial]

Utility score: 17.6

Best Action: 1

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 15.84

Best Action: 4

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 5.6

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 9.440000000000001

Best Action: 3

Location: V2

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 8.736

Best Action: 3

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 2.0

[Conf
ident
ial]

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

[Conf
ident
ial]

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

[Conf
ident
ial]

Location: V3

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 2.0

Best Action: 5

Location: V3

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 2.0

Best Action: 5

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 9.0

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 5.4

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

[Conf
ident
ial]

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 9.0

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 5.4

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 9.0

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 5.4

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 9.0

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

[Conf
ident
ial]

Utility score: 5.4

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 26.0

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 12.2

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 15.8

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 8.120000000000001

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 3.0

[Conf
ident
ial]

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 9.0

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 5.4

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 22.6

Best Action: 2

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 10.84

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 3.0

Best Action: 3

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 14.440000000000001

Best Action: 2

[Conf
ident
ial]

Location: V4

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 7.5760000000000005

Best Action: 3

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

[Conf
ident
ial]

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = FREE ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = BLOCKED ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

[Conf
ident
ial]

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = FREE ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = BLOCKED ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = FREE)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = BLOCKED)

Utility score: 0.0

Best Action: 0

Location: V5

Belief-State: ((V1,V3) = UNKNOWN ,(V2,V3) = UNKNOWN ,(V3,V4) = UNKNOWN)

Utility score: 0.0

Best Action: 0

===== Running a simulation #1 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V2, non-fragile edge. walking on it.

Agent location: 2 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #2 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #3 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V2, non-fragile edge. walking on it.

Agent location: 2 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #4 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V2, non-fragile edge. walking on it.

Agent location: 2 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 2 Best action: V4, non-fragile edge. walking on it.

Agent location: 4 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #5 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V2, non-fragile edge. walking on it.

Agent location: 2 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #6 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V2, non-fragile edge. walking on it.

Agent location: 2 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #7 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be blocked! transferring to the appropriate state...

Agent location: 1 Best action: V2, non-fragile edge. walking on it.

Agent location: 2 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #8 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #9 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

===== Running a simulation #10 =====

Agent starts an location V1 and all the fragile edges' status is unknown.

Agent location: 1 Best action: V3, fragile edge. It appears to be free, walking on it.

Agent location: 3 Best action: V5, non-fragile edge. walking on it.

Arrived at package location at V5

Finished the simulations... The program will now exit.